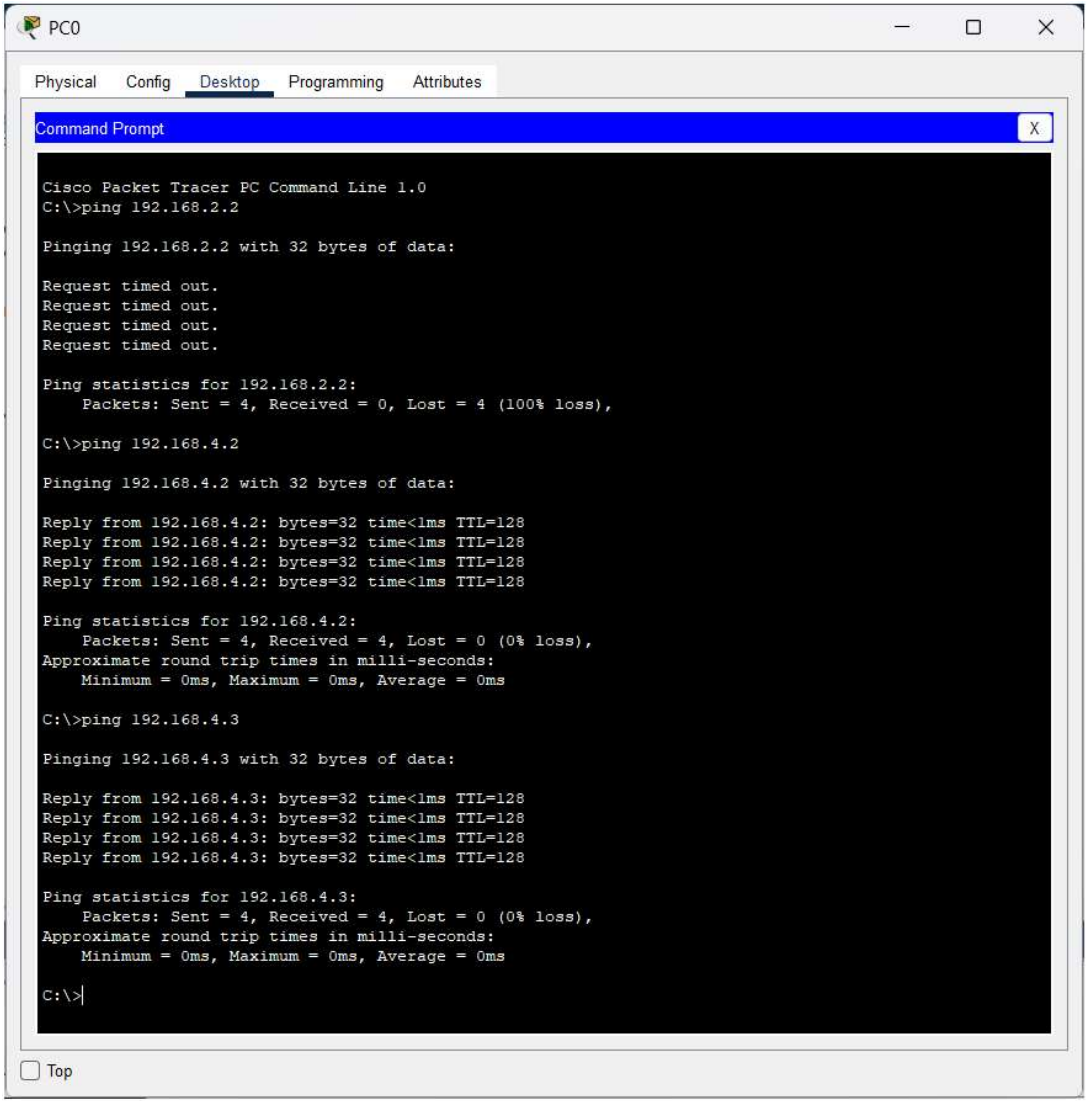
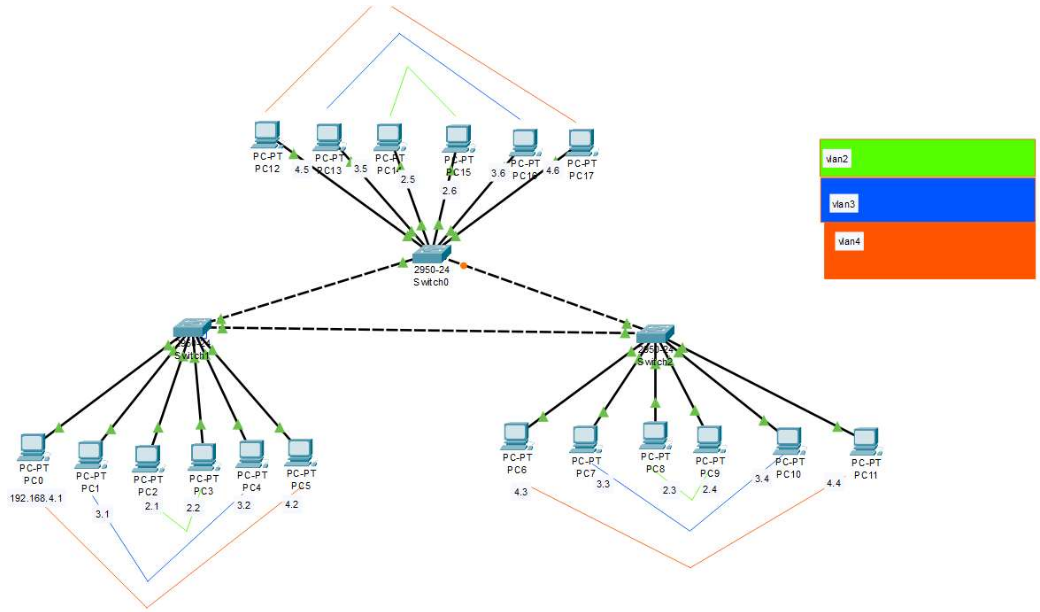
U24ai050 Lab-6

17 February 2026 17:51

1) Construct the below network in the Cisco packet tracer, as shown below, and connect them more securely using VLAN.





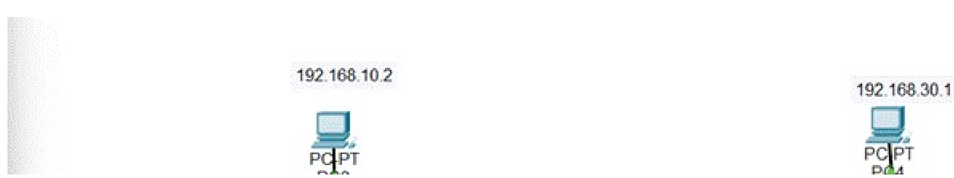
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC15	PC14	ICMP		0.000	N	13	(edit)	(delete)
	Successful	PC12	PC5	ICMP		0.000	N	14	(edit)	(delete)
	Successful	PC4	PC1	ICMP		0.000	N	15	(edit)	(delete)

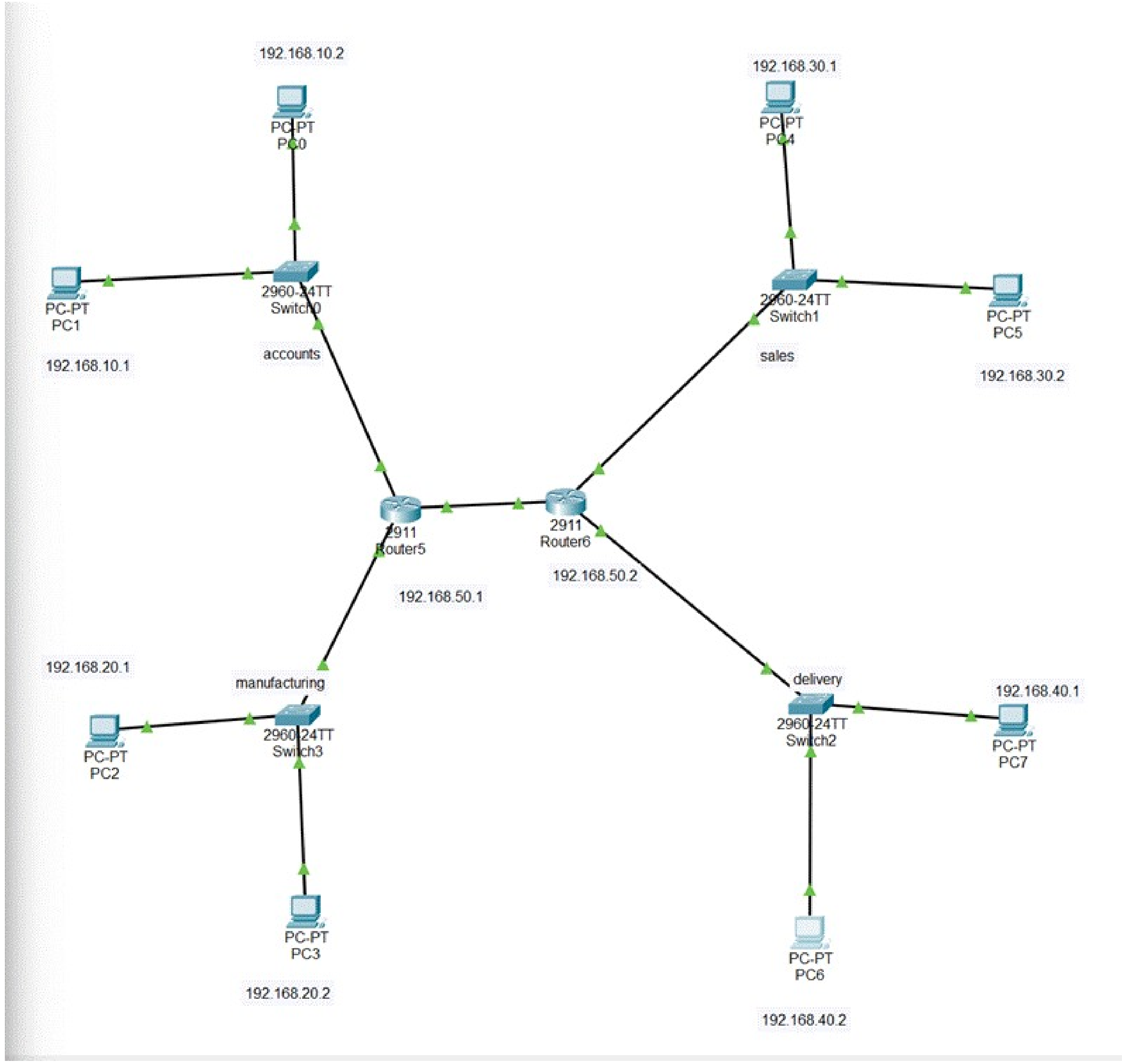
2) Design s network in CISCO packer tracer to connect Accounts, Sales, Delivery, and Manufacturing departments:

a) Each department should contain at least 2 PCs.

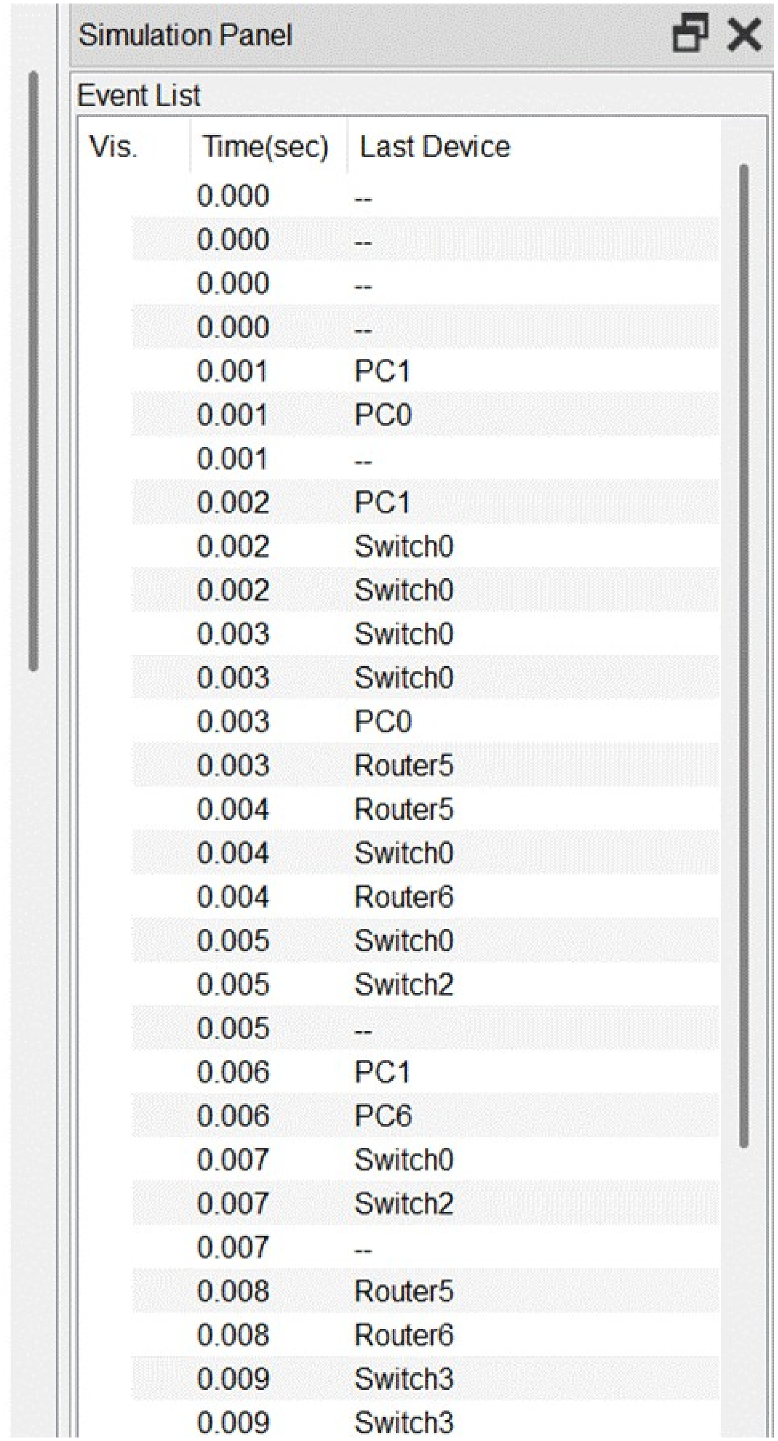
b) An appropriate number of switches and routers should be used in the network, and proper IP addresses with subnet masks should be given.

c) Connect all the devices with the appropriate wires and check their connectivity. d) Check the OSI layer information of the packet as shown below:




192.168.10.2
PC-PT
PC0
PC-PT
PC1
192.168.10.1
2960-24TT
Switch0
accounts
192.168.30.1
PC-PT
PC4
2960-24TT
Switch1
PC-PT
PC5
sales
192.168.30.2
2911
Router5
2911
Router6
192.168.50.2
192.168.50.1
192.168.20.1
manufacturing
PC-PT
PC2
2960-24TT
Switch3
PC-PT
PC3
192.168.20.2
delivery
192.168.40.1
2960-24TT
Switch2
PC-PT
PC7
PC-PT
PC6
192.168.40.2

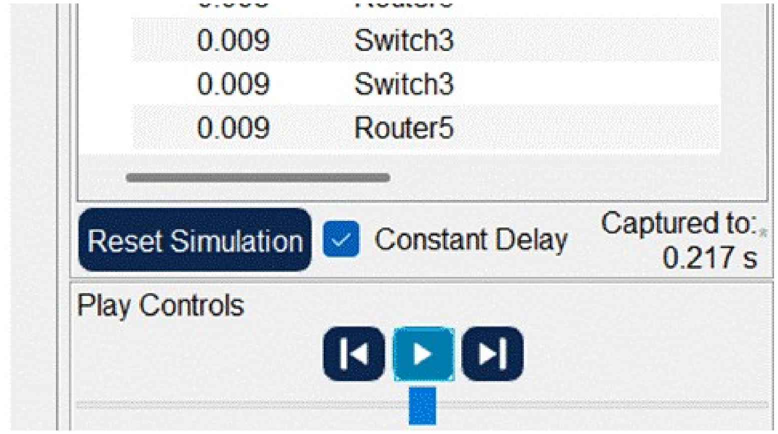

Simulation Panel

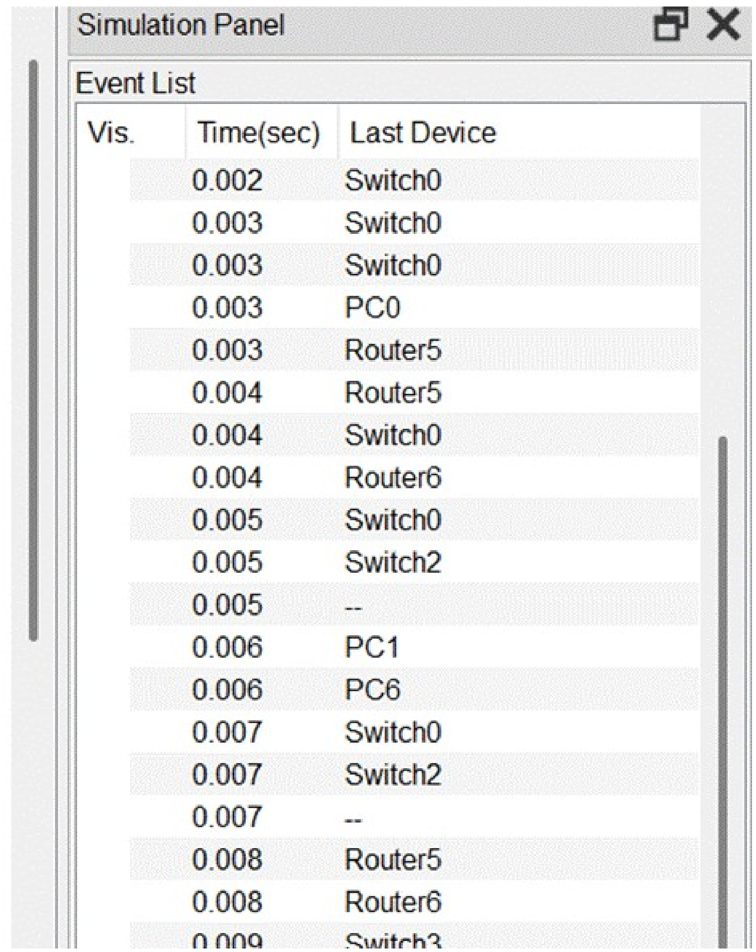


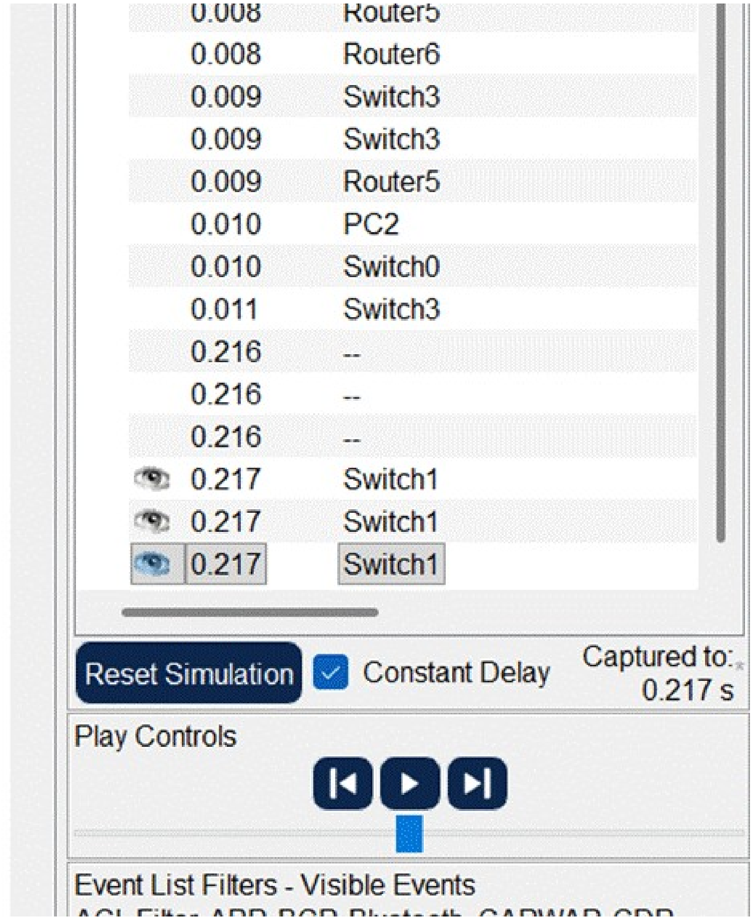
Simulation Panel

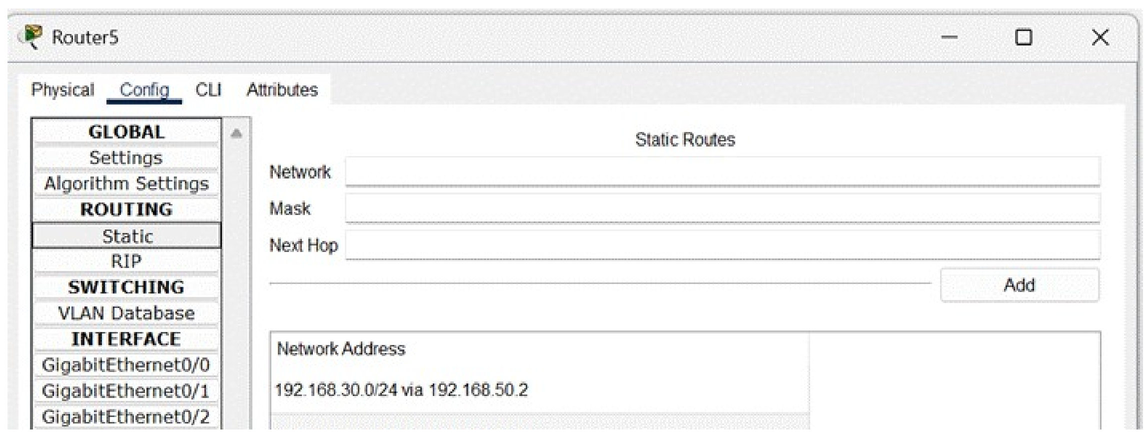
Event List

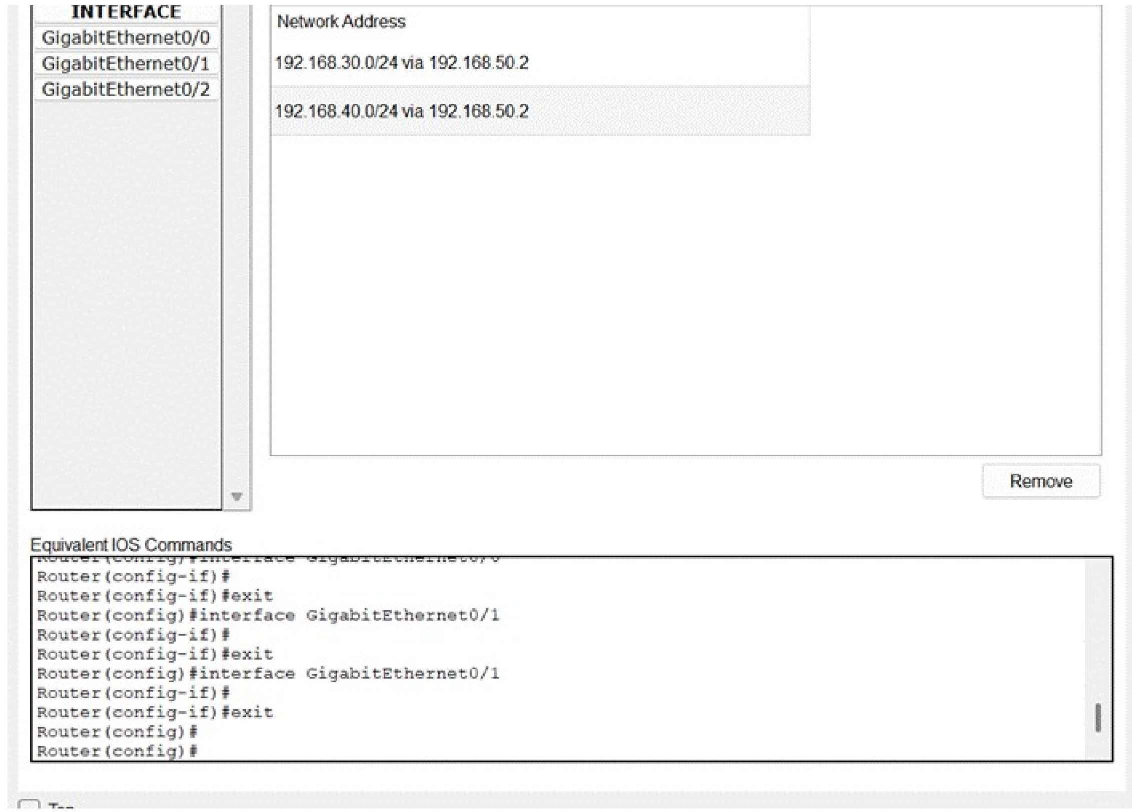
Vis.	Time(sec)	Last Device
	0.000	--
	0.000	--
	0.000	--
	0.000	--
	0.001	PC1
	0.001	PC0
	0.001	--
	0.002	PC1
	0.002	Switch0
	0.002	Switch0
	0.003	Switch0
	0.003	Switch0
	0.003	PC0
	0.003	Router5
	0.004	Router5
	0.004	Switch0
	0.004	Router6
	0.005	Switch0
	0.005	Switch2
	0.005	--
	0.006	PC1
	0.006	PC6
	0.007	Switch0
	0.007	Switch2
	0.007	--
	0.008	Router5
	0.008	Router6
	0.009	Switch3
	0.009	Switch3

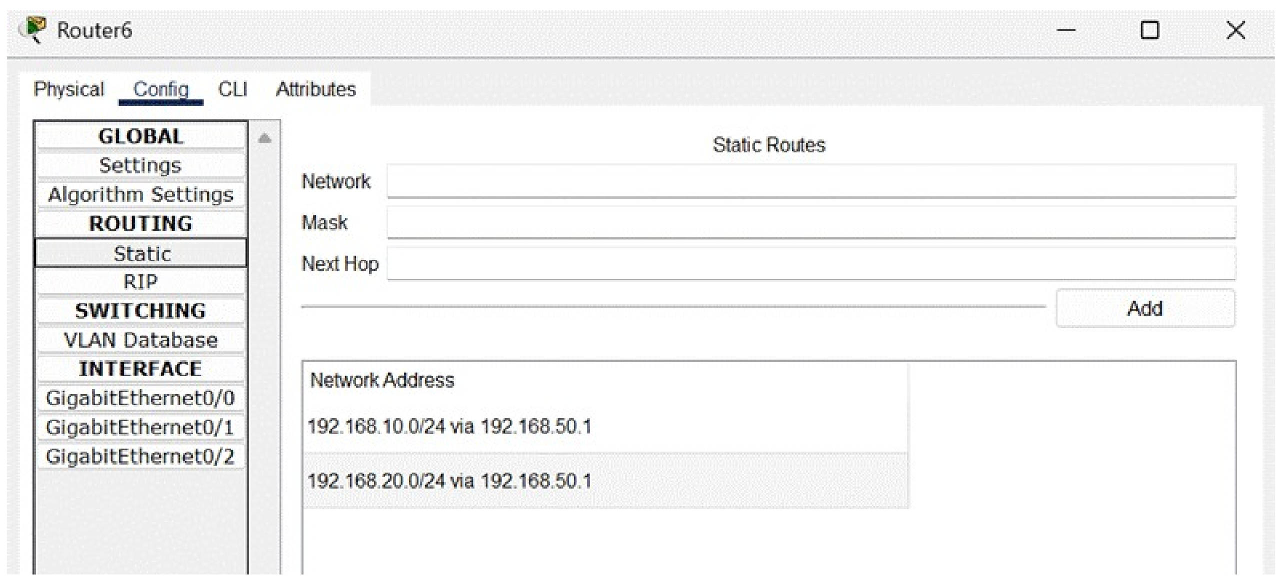

0.009 Switch3
0.009 Switch3
0.009 Router5
Reset Simulation
Constant Delay
Captured to:
0.217 s
Play Controls

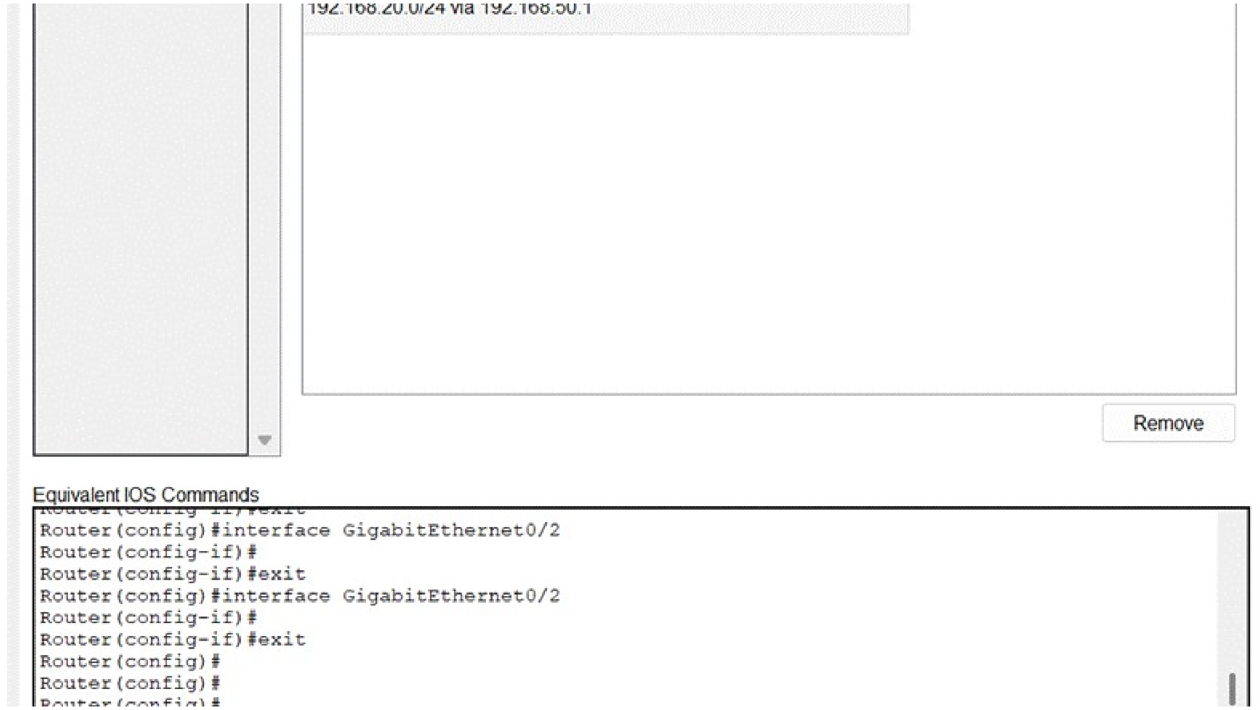

Simulation Panel
Event List
Vis. Time(sec) Last Device
0.002 Switch0
0.003 Switch0
0.003 Switch0
0.003 PC0
0.003 Router5
0.004 Router5
0.004 Switch0
0.004 Router6
0.005 Switch0
0.005 Switch2
0.005 --
0.006 PC1
0.006 PC6
0.007 Switch0
0.007 Switch2
0.007 --
0.008 Router5
0.008 Router6

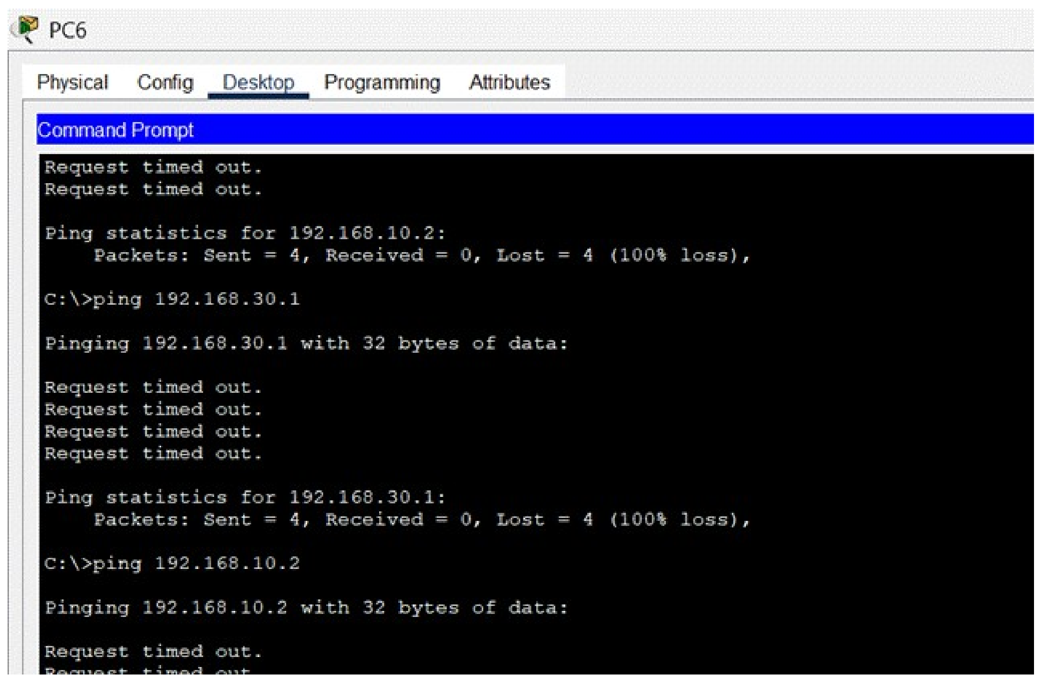

0.008 Router5
0.008 Router6
0.009 Switch3
0.009 Switch3
0.009 Router5
0.010 PC2
0.010 Switch0
0.011 Switch3
0.216 --
0.216 --
0.216 --
0.217 Switch1
0.217 Switch1
0.217 Switch1
Reset Simulation
Constant Delay
Captured to:
0.217 s
Play Controls
Event List Filters - Visible Events


Router5
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.30.0/24 via 192.168.50.2


INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
Network Address
192.168.30.0/24 via 192.168.50.2
192.168.40.0/24 via 192.168.50.2
Remove
Equivalent IOS Commands
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#


Router6
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.10.0/24 via 192.168.50.1
192.168.20.0/24 via 192.168.50.1


192.168.20.0/24 via 192.168.50.1
Remove
Equivalent IOS Commands
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#


PC6
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Request timed out.
Request timed out.
Ping statistics for 192.168.10.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.30.1
Pinging 192.168.30.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.30.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.2
Pinging 192.168.10.2 with 32 bytes of data:
Request timed out.

```
Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.10.2: bytes=32 time<1ms TTL=126
Reply from 192.168.10.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=126
Reply from 192.168.10.2: bytes=32 time<1ms TTL=126
Reply from 192.168.10.2: bytes=32 time<1ms TTL=126
Reply from 192.168.10.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.1: bytes=32 time<1ms TTL=127
Reply from 192.168.30.1: bytes=32 time<1ms TTL=127
Reply from 192.168.30.1: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=127
Reply from 192.168.30.1: bytes=32 time=1ms TTL=127
Reply from 192.168.30.1: bytes=32 time<1ms TTL=127
Reply from 192.168.30.1: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```